Credit Card Fraud Analysis Report

Introduction

Credit card fraud remains one of the most significant challenges facing financial institutions due to the increasing volume of digital transactions and the growing sophistication of cybercriminals. Fraudulent activities not only result in financial losses but also reduce customer trust and increase operational costs for banks and payment service providers. Data analytics has become an essential tool in identifying fraud patterns, assessing risks, and supporting informed decision-making.

Dashboard Overview

The dashboard summarizes credit card transaction activity through several Key Performance Indicators (KPIs) and interactive charts.

The dashboard reports:

- **Total Transactions:** 20,000
- **Fraud Rate:** 2%
- **Total Fraud Value:** 61.44K
- **Average Legitimate Transaction Amount:** 131.58
- **Average Fraudulent Transaction Amount:** 181.25

These indicators provide a quick overview of the organization's fraud exposure. Although the fraud rate is relatively low at approximately 2%, fraudulent transactions represent significant financial losses because they involve higher transaction values than legitimate purchases.

Analysis and Findings

1.Fraud Rate by Card Type

The dashboard compares fraud rates among different card providers, including Amex, Discover, Visa, Mastercard, and RuPay. The analysis shows that **Amex** has the highest fraud rate, followed closely by **Discover** and **Visa**, while **RuPay** records the lowest fraud rate.

Premium credit cards often become attractive targets because they generally have higher spending limits and are associated with larger transaction amounts. This insight suggests that financial institutions should strengthen fraud monitoring for higher-risk card categories while maintaining robust security measures across all card types.

2.Fraud Rate by Authentication Method

Authentication methods play a critical role in reducing fraud. The dashboard indicates that transactions completed **without authentication** have the highest fraud rate. Fraud levels decrease considerably when security mechanisms such as **One-Time Passwords (OTP)**, **PIN verification**, and **3D Secure** are used. **Biometric authentication** records the lowest fraud rate, making it the most effective authentication method displayed in the dashboard. These findings demonstrate that stronger authentication significantly improves transaction security and reduces fraudulent activities.

3.Fraud by Transaction Channel and Device

The dashboard evaluates fraud across multiple transaction channels, including ATM, Contactless, Online, In-App, and Point of Sale (POS), while also comparing different devices such as Android phones, iPhones, Mac computers, ATM machines, and POS terminals. The analysis shows relatively consistent fraud rates of approximately **2%** across all channels and devices. This indicates that fraud is widespread rather than concentrated within one specific payment method or device. Organizations should therefore implement fraud detection systems consistently across all transaction channels instead of focusing solely on online or mobile transactions.

4.Foreign vs Domestic Fraud

One of the most significant findings is the comparison between domestic and foreign fraudulent transactions. The dashboard shows that approximately **80%** of fraud cases originate from foreign transactions, while only about **20%** occur domestically. This suggests that international transactions present a much higher fraud risk. Banks should therefore apply additional security measures such as transaction monitoring, customer verification, and risk scoring for overseas purchases.

5.AI Scam Fraud Analysis

The dashboard also categorizes fraud into AI-assisted scams and traditional fraud. The analysis indicates that **AI-related scams account for approximately 85%** of all fraud cases, while only about **15%** are classified as non-AI scams. This trend reflects the increasing use of artificial intelligence by cybercriminals to automate phishing attacks, social engineering, and fraudulent transaction attempts. Financial institutions should invest in AI-powered fraud detection systems capable of identifying evolving fraud patterns in real time.

6.Fraud Cases by Time of Day

Time-based analysis provides valuable operational insights. According to the dashboard, the **Early Morning** period records the highest number of fraudulent transactions, while Morning, Afternoon, and Evening experience significantly fewer fraud cases. This pattern suggests that fraudsters may exploit periods when customers are less active or when monitoring activities are reduced. Increasing fraud surveillance during these high-risk hours can improve fraud detection and reduce financial losses.

Business Insights

The dashboard highlights several important insights that can support business decision-making.

First, although fraudulent transactions account for only a small percentage of all transactions, they involve significantly higher transaction values than legitimate purchases. This increases the financial impact of fraud despite its relatively low frequency.

Second, stronger authentication methods such as biometric verification and OTP significantly reduce fraud risk compared to transactions without authentication.

Third, foreign transactions represent the largest source of fraud, suggesting that international payment activity should receive additional monitoring.

Finally, the growing dominance of AI-assisted scams demonstrates the need for financial institutions to adopt advanced technologies capable of detecting sophisticated fraud techniques.

Recommendations

Based on the analysis, the following recommendations are proposed:

- Implement mandatory Multi-Factor Authentication (MFA) for high-value and high-risk transactions.
- Increase monitoring of international transactions using real-time fraud detection systems.
- Deploy Artificial Intelligence and Machine Learning models to identify emerging fraud patterns.
- Apply additional verification procedures for premium cardholders and higher-value transactions.
- Increase fraud monitoring during early morning hours when fraudulent activity is highest.
- Conduct regular customer awareness campaigns on phishing, identity theft, and AI-assisted scams.
- Continuously review fraud trends by card type, authentication method, and transaction channel to improve fraud prevention strategies.

Conclusion

The Credit Card Fraud Analysis Dashboard demonstrates how business intelligence tools such as Microsoft Power BI can transform transactional data into actionable insights. Despite an overall fraud rate of only **2%**, fraudulent transactions result in substantial financial losses due to their higher average value. The analysis identifies several key risk areas, including premium card types, transactions without authentication, international transactions, AI-assisted scams, and early morning transaction activity by implementing stronger authentication mechanisms, leveraging AI-powered fraud detection, and enhancing monitoring during high-risk periods, financial institutions can significantly reduce fraud exposure while improving customer confidence and operational efficiency. This project illustrates the value of data analytics in supporting strategic decision-making and strengthening fraud prevention in the financial services industry.